



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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Class : 12

UNIT TEST - 2, AUGUST - 2024

Time Allowed : 1.30 Hours]

CHEMISTRY

[Max. Marks : 35

PART - I

(i) Answer all the questions.

(ii) Choose the correct answer.

5x1=5

- Which of the following is strongest acid among all?
(a) HI (b) HF (c) HBr (d) HCl
- What is the oxidation state of Phosphorous in Hypophosphorous acid [H_3PO_2]
a) +1 b) +2 c) +3 d) +4
- Which of the following Lanthanoid ion is diamagnetic?
a) Eu^{2+} b) Yb^{2+} c) Ce^{2+} d) Sm^{2+}
- If the solubility product of lead iodide is 3.2×10^{-8} . Its solubility will be
a) $2 \times 10^{-3} \text{ M}$ b) $4 \times 10^{-4} \text{ M}$ c) $1.6 \times 10^{-5} \text{ M}$ d) $1.8 \times 10^{-5} \text{ M}$
- Assertion** : P - N, N - dimethyl amino benzaldehyde undergoes benzoin condensation
Reason : The Aldehyde (-CHO) group is meta directing
(a) If both Assertion and Reason are true and reason is the correct explanation of Assertion.
(b) If both Assertion and Reason are true but reason is not the correct explanation of Assertion
(c) Assertion is true but Reason is False (d) Both Assertion and Reason are False

PART - II

II. Answer any three questions. Q.No. 9 is compulsory.

3x2=6

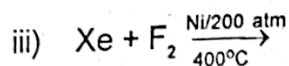
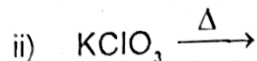
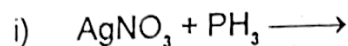
- Give the uses of Helium.
- What are Inner Transition Elements?
- Define Solubility Product.
- A Solution of 0.10 M of a weak electrolyte is found to be dissociated to the extent of 1.20% at 25°C. Find the dissociation constant of acid.
- Write the Structure of the major product of the aldol condensation of Benzaldehyde with Acetone.



III. Answer any three questions. Q. no. 15 is compulsory.

3

11. Complete the following Reactions?



12. Justify the position of Lanthanoids and Actinoids in the Periodic Table.

13. Out of $\text{Lu}(\text{OH})_3$ and $\text{La}(\text{OH})_3$, which is more basic and Why?

14. Derive the Expression for Ostwald's Dilution law.

15. Explain Rosenmund Reduction Reaction.

PART - IV

IV. Answer all the questions.

3x5

16. a) i) How will you prepare Chlorine in Laboratory. (3)

ii) Write the reason for the Anomalous behaviour of Nitrogen. (2)

(OR)

b) What is Lanthoid Contraction and What are effects of Lanthanoid Contraction. (5)

17. a) i) What are Lewis Acids and Bases. Give two example for each. (3)

(ii) Discuss the Lowry - Bronsted concept of Acid and Bases. (2)

(OR)

b) i) Define P^H . (2)

ii) Differentiate White Phosphorous and Red Phosphorous. (3)

18. a) i) How will you prepare Urotropine? Draw the Structure of Urotropine. (3)

ii) Write the uses of Formaldehyde. (2)

(OR)

b) i) Calculate the Magnetic moment value for Mn^{2+} ion (3)

ii) Explain the Catalytic properties of Transition Elements. (2)